DAYANANDA SAGAR UNIVERSITY, BENGALURU

School of Engineering • End Semester Examination

Course: Linear Algebra and Calculus | Course Code: 25EN1101 | Credits: 4

Exam Session: End Semester Exam 2024 **Max Marks: 80 | Duration: 3 Hours**

INSTRUCTIONS TO CANDIDATES:

1. Answer FIVE full questions, choosing ONE full question from each module.
2. Draw neat diagrams wherever necessary. Missing data may be suitably assumed.

MODULE 1

Q1. Find the Eigenvalues and corresponding Eigenvectors for the matrix A = [[1, 2, 2], [2, 1, 2], [2, 2, 1]]. Verify the Cayley-Hamilton Theorem and find A^-1. **[8 Marks]**

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Q2. State and prove Euler's Theorem for a homogeneous function u = f(x, y) of degree n. Hence evaluate x(du/dx) + y(du/dy) for u = sin^-1((x^2 + y^2)/(x + y)). **[8 Marks]**

MODULE 2

Q3. Expand f(x, y) = e^x * cos(y) in powers of (x - 1) and (y - pi/4) up to second-degree terms using Taylor's Theorem for two variables. **[8 Marks]**